

Theory of Computation (CS2030) : Problem Set 2

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Please read the following comment before you work on the problems.

This is a set of questions that you should solve to master the contents taught in the class. *You don't have to submit the solutions!* But you should still solve them to understand the material, and also because these problems or their variants have a devilish way of finding their way in exams/quizzes!

We suggest to try to solve them from scratch under exam conditions—by yourself, *without* your notes, *without* the internet, and if possible, even without a cheat-sheet.

If you find yourself getting stuck on a problem, try to figure out *why* you're stuck. Do you understand the problem statement? Are you stuck on choosing the right high-level approach, or are you stuck on the details? Are you solving the right subproblems?

Discussing problems with other people (in your study groups, with me, on Google Classroom, or with LLMs) and/or looking up old solutions can be *extremely* helpful, but **only after** you have (1) made a good-faith effort to solve the problem on your own, and (2) you have either a candidate solution or some idea about where you are getting stuck.

When you do discuss problems with other people, remember that your goal is not merely to “understand” the solution to any particular problem, but to become more comfortable with solving a certain type of problem on your own. Identify specific steps that you find problematic, read more about those steps, and focus your practice on those steps.

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- Let $A = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and if } i = 1 \text{ then } j = k\}$. Show that A is not regular.
 - Show that A acts like a regular language in the pumping lemma. In other words, give a pumping length p and demonstrate that A satisfies the three conditions of the pumping lemma for this value of p .
 - Let M_1 and M_2 be two DFAs given as input. Give a polynomial-time algorithm to decide whether the two automata accept the same language, i.e., $L(M_1) = L(M_2)$.
 - Let $A = \{0^i 1^j \mid \gcd(i, j) = 1\}$. Show that A is not regular.
 - Let $A = \{0^i 1^j \mid i > j\}$. Show that A is not regular.
 - Let $A = \{0^n 1^n 2^n \mid n \geq 0\}$. Show that A is not regular.
 - Let $A = \{www \mid w \in \{a, b\}^*\}$. Show that A is not regular.
 - Let $A = \{0^i 1^j \mid i \neq j\}$. Show that A is not regular.

8. Let $A = \{0^i 1^j 0^i \mid i, j \geq 0\}$. Show that A is not regular.
9. Let $A = \{w \in \{0,1\}^* \mid w \text{ is not a palindrome}\}$. Show that A is not regular.
10. Let $\Sigma = \{a, b\}$. For each $k > 1$, let C_k be the language consisting of all strings that contain an a exactly k places from the right-hand end. Thus $C_k = \Sigma^* a \Sigma^{k-1}$. Describe an NFA with $k + 1$ states that recognizes C_k . Further, Prove that for each k , no DFA can recognize C_k with fewer than 2^k states.
11. Let A be a regular set. Consider the two sets
 - (a) $\{x \mid \exists n \geq 0, \exists y \in A, \text{ such that } y = x^n\}$
 - (b) $\{x \mid \exists n \geq 0, \exists y \in A, \text{ such that } x = y^n\}$

One is necessarily regular and one is not. Which is which? Give a proof and a counterexample.

12. Let p be any polynomial of degree d with nonnegative integer coefficients. Show that if A is a regular set, then so is the set $A' = \{x \mid \exists y \text{ such that } |y| = p(|x|) \text{ and } y \in A\}$.